impenetrable in this tongue. Much insight into some of the latter passages might emerge from the Inuit translation; in yet other sections that translation might be completely opaque. Sanskrit might capture the essence of some of these tricky passages, but for other, particularly troublesome sections, all five translations might leave you dumbfounded and only the master text will be intelligible. This is much closer to the situation with the five string theories. Theorists have found that for certain questions, one of the five may give a transparent description of the physical implications, while the descriptions given by the other four are too mathematically complex to be useful. And therein lies the power of Witten's discovery. Prior to his breakthrough, string theory researchers who encountered intractably difficult equations would be stuck. But Witten's work showed that each such question admits four mathematical translations—four mathematical reformulations—and sometimes one of the reformulated questions proves far simpler to answer. Thus, *the dictionary for translating between the five theories can sometimes provide a means for translating impossibly difficult questions into comparatively simple ones.*

It's not foolproof. Just as all five translations of certain passages in that master text might be equally incomprehensible, sometimes the mathematical descriptions given by all five string theories are equally difficult to understand. In such cases, just as we would need to consult the original text itself, we would need full comprehension of the elusive M-theory to make progress. Even so, in a wealth of circumstances, Witten's dictionary has provided a powerful new tool for analyzing string theory.

Hence, just as each translation of a complex text serves an important purpose, each string formulation does too. By combining insights gained from the perspective of each, we are able to answer questions and reveal features that are completely beyond the reach of any single string formulation. Witten's discovery thus gave theorists five times the firepower for advancing string theory's front line. And that, in large part, is why it sparked a revolution.

Eleven Dimensions